

VOICE GUARD

Telling a real human voice apart from an AI-generated one.



01 OVERVIEW

Voice Guard is a deep-learning system that listens to a short speech clip and decides whether it was **spoken by a real person** or **generated by AI** (voice cloning or text-to-speech).

Cheap, high-quality voice synthesis has made cloned-voice fraud, impersonation, and misinformation a real and growing threat. At its core the task is a binary classification problem — real vs. fake — but the hard, interesting part is **generalization**: most detectors do well on the synthesis methods they were trained on and fail on new, unseen ones. Voice Guard treats that gap as the central problem, not an afterthought.

02 OBJECTIVES

- Build a reliable baseline detector that classifies speech as real or AI-generated on a standard benchmark.
- Fine-tune a pretrained speech model (Wav2Vec 2.0 / XLS-R) to reach strong detection accuracy.
- Measure and improve **generalization** to synthesis methods the model has never seen — the project's core contribution.
- Report results with **EER (Equal Error Rate)**, the standard metric in the field, rather than accuracy alone.
- Deliver a working demo that takes an uploaded clip and returns a verdict with a confidence score.

- A** **Data preparation.** Load the benchmark, parse its protocol files, and convert each audio clip into a Mel-spectrogram representation the model can learn from.
- B** **Baseline model.** Train a small CNN on the spectrograms to validate the full pipeline end-to-end. Expected accuracy is modest — the goal is a working loop.
- C** **Main model.** Replace the CNN with a pretrained Wav2Vec 2.0 / XLS-R feature extractor plus a lightweight classification head; fine-tune for a clear jump in performance.
- D** **Generalization test.** Train on one dataset and evaluate on a different, real-world one (In-the-Wild). The performance drop is the finding the project is built around.
- E** **Robustness improvements.** Apply data augmentation — background noise and codec/compression effects — to narrow the generalization gap, comparing EER before and after.
- F** **Demo & evaluation.** Wrap the model in a simple interface (Gradio / Streamlit) for live testing, and compile EER result tables across all datasets.

04 DATA & SOURCES

DATASET	ROLE	WHAT IT CONTAINS
ASVspoof 2019 LA	Training	~25k clips; fake speech from 19 different text-to-speech and voice-conversion systems. The standard starting benchmark.
ASVspoof 2021 DF	Evaluation	High-fidelity deepfakes passed through lossy codecs — a tougher, more realistic test set.
In-the-Wild	Generalization test	Real-world deepfakes of public figures collected from the internet — the key set for measuring true generalization.
ASVspoof 5	Optional / stretch	The newest and largest crowdsourced benchmark (2024) for scaling up if time allows.

Where to get them. The ASVspoof datasets are published on the official site (asvspoof.org) and mirrored on Edinburgh DataShare. The In-the-Wild dataset is released by Fraunhofer AISEC on its deepfake evaluation page. Many of these are also mirrored on Kaggle and Hugging Face Datasets for faster download. All are free for research use — check each dataset's license before publishing results.

Python

PyTorch

torchaudio

Hugging Face Transformers

Wav2Vec 2.0 / XLS-R

Gradio / Streamlit

Metric: EER

Success is not "95% accuracy." It is a clear comparison of performance on *seen* vs. *unseen* synthesis methods — showing the problem was understood, not just a library run.